

Task Manager App

A calendar-centric life dashboard: tasks, fitness, medication, payments, and expenses in one place



What's new in v3:

- User / Admin / Caregiver role architecture
- Offline handling & real-time offline detection
- Timezone handling for reminders and escalation
- Full revocation & teardown model, incl. autonomy safeguards
- Caregiver reassignment flow
- Full app flow — 6 diagrams end to end (Section 16)

1. Core concept

The app is a calendar-centric task manager. The **Home screen is a live calendar**, and four separate life-domain modules feed into it: **Fitness**, **Reminders** (Pill, Payment, Appointment, Custom), and **Expense**. The calendar is strictly an **aggregator and preview layer** — it never owns or duplicates module data.

2. Navigation structure

Tab	Role
Home	Calendar view. Aggregates and previews entries from all other modules.
Fitness	Workout logs, goals, and AI-assisted plan generation.
Reminders	Pill, Payment, Appointment, and Custom — grouped under one tab, each with its own icon.
Expense	Categorized spending ledger; rolls up into daily, weekly, and monthly totals.

3. The "+" add flow

Module-first, always. Global "+" shows module picker, then that module's categories. Tapping a specific date on Home shows module options directly with the date pre-filled.

4. Reminders module

4.1 Pill MVP: manual entry v2: auto-lookup

Functions as a lightweight medical ID for elderly users. **Fields:** name, dosage, time(s)/day, expiry, photo.

4.2 Payment / Bill

Fields: payee, amount, due date, recurring flag. Links into Expense — see Section 6.3.

4.3 Appointment

Kept deliberately simple — title, location, time only.

4.4 Custom

User-picked emoji represents a freeform task. Same-day reuse of an emoji groups tasks under that bucket (distinct timestamps required) rather than duplicating entries; the calendar shows a count badge. Tracking is set per item as streak/checklist or numeric.

4.5 Timing & recurrence

Timing is module-specific (Pill = times/day, Payment = days-before-due, Fitness/Custom = simple time). Recurrence offers quick presets (next day, 2/3 days after, same day next week/month) plus full custom flexibility for irregular patterns.

4.6 Missed / overdue — escalation ladder

Loud by default, with a per-user toggle to quiet it down. Missed pills specifically stay visible to a linked caregiver.

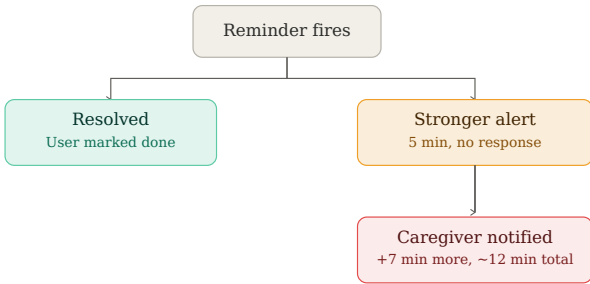


Fig 1 — Missed reminder escalation flow

5. Fitness module

Objects: Workout log (retrospective) and Goal (prospective). Goals drive the calendar forward — planned days appear in advance.

Plan creation, two paths: an AI-assisted chatbox that gathers fitness level/goal/equipment/days and proactively drafts a plan (user refines after), or fully manual entry. Diet-plan generation was considered and removed — workouts only.

Plan vs. actual: once a user edits an AI-suggested workout, the edited version becomes the real plan. Actual logged results are compared against that edited plan for goal/streak tracking.

v2 Wearable / step-counter sync — MVP is manual logging only.

6. Expense module

Categories: Food, Transport, Bills/Utilities, Shopping, Health, Other. Tapping a date shows that day's spend; an end-of-month section shows weekly breakdowns.

Payment ↔ Expense link: a paid Payment auto-creates a linked Expense entry (Bills), with a per-payment opt-out toggle. Paid and unpaid both count toward the monthly total, visually distinguished.

7. Home calendar — visual system

Month view is the default. Dates are shown as density-coded shapes, not icons:

Task count	Color	Edge style
0	Translucent / neutral	—
1-4	Green	Soft, rounded
5-7	Orange	Medium
8+	Red	Rough, sharper

Edge shape is a deliberate accessibility layer alongside color. No numbers or icons live inside grid cells — daily/monthly spend lives behind tap-in detail and the end-of-month summary section.

8. Onboarding & signup

Signup collects name, age, location permission, phone, and email, with a separate setup path for "for a child" or "for an elderly person." New users are forced through **Pill + Payment** (fixed, highest-consequence) before a skip option appears; which modules are forced can vary by profile risk level, and elderly/45+ profiles are forced through all 4. The walkthrough offers real data entry or a pre-recorded demo video.

9. App architecture — User / Admin / Caregiver roles

Role	Definition
User	The person themselves — an elderly person using their own account and a regular user are the same role.
Admin	The role held by someone overseeing another person's account. Holds overall authority — including assigning and reassigning who acts as Caregiver.
Caregiver	A role the Admin assigns (to themselves or to someone else, e.g. a hired nurse) specifically for day-to-day monitoring.

Open: Admin holds overall authority, but the User can take back control in certain ways when needed — the exact scope of this override is not yet fully specified.

10. Caregiver monitoring

10.1 Setup & visibility

- Caregiver access requires **mutual consent** from both people to begin — never unilateral.
- The caregiver sees **full detail** (medication names, dosages, expiry), not just a status signal.
- The monitored person can always see that they are being watched — monitoring is never silent.

10.2 Revocation & teardown

- The **caregiver** can revoke access at any time.
- The **elderly/monitored person** can also revoke access independently — either via a 6-digit code sent to the caregiver's phone (the mutual/routine path), or by stating a reason in writing via mail/email (the one-sided path, usable without the caregiver's cooperation).
- A child profile above 18 can revoke access themselves; a child profile below 18 cannot revoke access once granted.
- When access is revoked, it ends **immediately** and all data is wiped from the former caregiver's view — no lingering historical access.

Autonomy safeguard: giving the monitored person an independent, caregiver-cooperation-free revocation path (the written-reason route) prevents monitoring from becoming a one-way, permanent loss of control — a real risk this design deliberately closes off.

10.3 Caregiver reassignment

If a caregiver quits, the Admin can assign a new one. Historical data is preserved through the transition rather than lost.

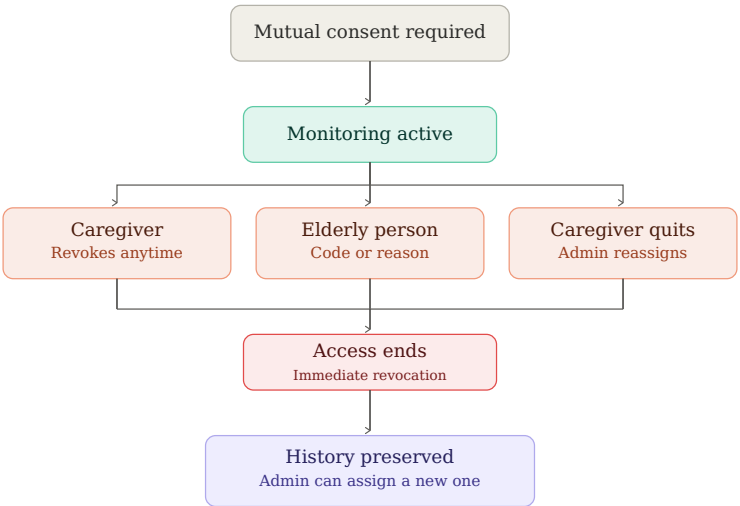


Fig 2 — Caregiver access lifecycle

11. Offline & connectivity handling

- Reminders fire **locally** on the monitored person's phone even with no network — the schedule doesn't depend on connectivity.
- Offline detection is **real-time**: the moment a monitored phone goes offline, the caregiver is notified on the spot — not retroactively after reconnection.
- The caregiver can see scheduled/upcoming reminders in real time (pre-known data), but cannot see completion status (done / not done) until connectivity is restored, at which point all queued data syncs at once.

12. Timezone handling

Reminders and escalation timing run entirely on the **monitored person's own timezone** — never the caregiver's. The caregiver's app only needs to display timestamps intelligibly across timezones; the underlying schedule and escalation clock are unaffected by where the caregiver is.

13. Open risks & unresolved items

- User override scope.** Admin holds overall account authority, but the User can reportedly take back control in certain ways — exact mechanics not yet defined.
- Elderly-simplicity vs. system depth.** Needs a UX pass once real screens exist to confirm the simple path (add a pill, view calendar, get reminded) never requires touching AI plans, emoji grouping, or multi-role account management.
- Notification delivery engine.** Escalation timing is defined; the underlying push infrastructure to deliver it reliably is still a build item.

14. Deferred / v2 scope

Item	Why deferred
Pill drug-database auto-lookup	Requires integrating an external medicine database.
Fitness wearable / step-counter sync	Real integration project (device APIs).
Full multi-role permission system	Core rules are defined (Section 9-10); the complete permission engine is bigger than a first release.

15. Still to design

- User-override mechanics (Section 13).
- Visual mockups for each module's detail screens.
- Data security / encryption approach (engineering-level, not IA, but must not be forgotten given the medical and financial data involved).

16. Full app flow — diagrams

Six flows covering the app end to end: how someone enters the app, how any entry gets created, how Fitness plans get built and reconciled, how Payments link to Expense, and the two lifecycle flows already shown in Sections 4 and 10 (repeated in the list below for completeness).

16.1 Entry flow — signup to Home

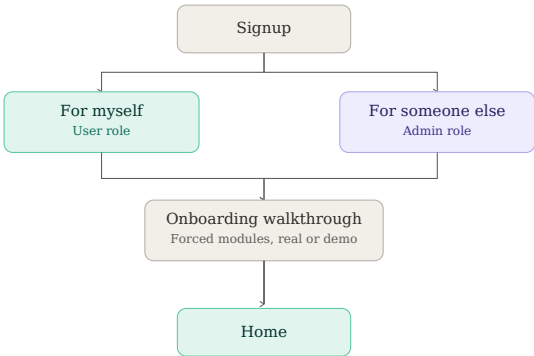


Fig 3 — App entry flow

16.2 Add-entry flow — any module, any starting point

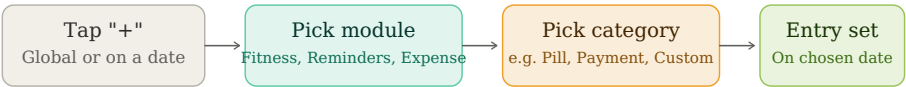


Fig 4 — Add-entry flow

16.3 Fitness plan flow

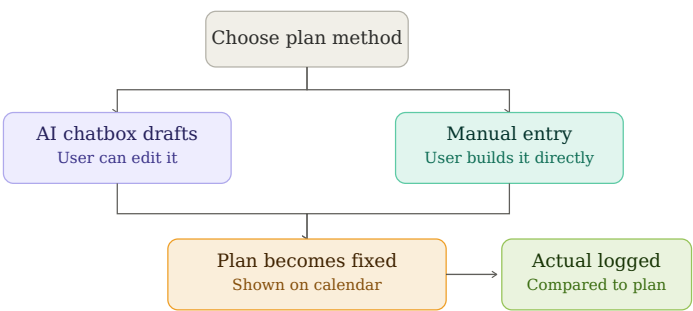


Fig 5 — Fitness plan flow

16.4 Payment to Expense link flow



Fig 6 — Payment to Expense link flow

16.5 Already shown — Reminders and Caregiver lifecycles

Fig 1 (Section 4.6) — the missed reminder escalation flow: reminder fires, resolves or escalates through a stronger alert into a caregiver distress notification.

Fig 2 (Section 10.3) — the caregiver access lifecycle: mutual consent to start, active monitoring, three ways access can end, and history preservation for reassignment.